

Zachary Deutsch

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Education

DUKE UNIVERSITY

Durham, NC

B.S. Mathematics; B.S.E. Mechanical Engineering; Minor in Financial Economics; Minor in Philosophy; GPA: 3.7 August 2024 – May 2028
Coursework: Calculus I–III; Linear Algebra; Differential Equations; Probability; Statistics & Data Analysis; Data Structures & Algorithms

UNIVERSITY OF OXFORD

Oxford, England

Study Abroad: Philosophy & Ethics of AI

June – August 2025

Experience

DUKE UNIVERSITY, Oca Lab

Durham, NC

Data Analyst

October 2025 – Present

- Benchmarked LLM hallucination rates in reasoning tasks using Monte Carlo simulation to estimate error distributions and tail risk

DUKE UNIVERSITY, Hickey Lab

Durham, NC

Quantitative Analyst

November 2024 – Present

- Designed and benchmarked unsupervised clustering pipelines on ~1M high-dimensional spatial-omics observations, improving signal quality and classification accuracy under noisy data conditions

TEL AVIV UNIVERSITY, Center for AI and Data Science

Tel Aviv, Israel

Quantitative Research Intern

May – August 2025

- Evaluated quantile-based feature extraction for time-series classification, benchmarking predictive performance across multiple classifiers

SHIRLEY RYAN ABILITYLAB

Chicago, IL

Data Analyst Intern

June – August 2024

- Analyzed outcomes from a \$1.5M DoD-funded study using statistical analysis and visualization to support performance evaluation and data-driven decision-making

Publications

- Deutsch, Z.**, Staupé R. & Bartoszewicz M. G. (forthcoming). *AI as a catastrophic imaginary*. Routledge Handbook of Catastrophic Imaginaries.
- Deutsch, Z.**, Surguladze, N., Miao, Y., & Hickey, J. W. (forthcoming). *Evaluation of clustering methods and large language models for CODEX cell type annotation*. EXO Beyond the Cell.
- Deutsch, Z.**, Eva, R., & Oca, S. R. (2026). *Virtue ethics and student use of large language models in engineering education*. IEEE Frontiers in Education Conference.
- Deutsch, Z.**, & Oca, S. R. (2026). *Evaluating engineering students' detection of hallucinations in large language model solutions*. American Society for Engineering Education Annual Conference.
- Deutsch, Z.** (2025). *Superintelligence, Instrumental Convergence, and the Limits of AI Apocalypse*. AI and Ethics (Springer Nature).

Skills, Awards, & Interests

Technical: Python (NumPy, pandas); SQL; APIs; Excel (financial modeling, analysis); PowerPoint

Languages: Japanese (Limited Working Proficiency); Spanish (Limited Working Proficiency)

Awards: 1st Place J.P. Morgan GenerationTech, National Youth Science Academy, Google Scholarship

Interests: Espresso, Liszt, Stoicism, Nietzsche, Cycling